

## LX200 Classic Command Set

Intended for professional programmers, the LX200 Command Set is used to write custom software for remote operation of the telescope with a personal computer. Each command is listed in a section appropriate to its type. Each entry in the command list includes the command name and any parameters which must be appended, any values which might be returned and a description of the command. The parameters and the return data are shown in a manner that indicates their format. These formats are listed below along with examples of how the data might actually appear, the allowable range of values and a short description.

This command set is presented here as it appears in the Instruction Manual dated 1996 Ver 0196-2

### a. Command Set Formats

HH:MM.T Example: 05:47.4

Range: 00:00.0 to 23:59.9  
Hours, minutes, and tenths of minutes.

sDD\*MM Example: +45\*59

Range: -90\*00 to +90\*00  
Signed degrees and minutes (the (\*) represents ASCII 223 which appears on the hand controller as a degree symbol).

DDD\*MM Example: 254\*09

Range: 000\*00 to 359\*59  
Unsigned degrees and minutes.

HH:MM:SS Example: 13:15:36

Range: 00:00:00 to 23:59:59  
Hours, minutes, and seconds.

MM/DD/YY Example: 02/06/92

Range: 01/01/00 to 12/31/99 (see description)  
Month, day, and year. The two digit year indicates the following: 92 through 99 = 1992 through 1999 and 00 through 91 = 2000 through 2091

sHH Example: -5 Range: -24 to +24  
Signed hour offset.

NNNN Example: 3456

Range: 0000 to 9999  
Four digit object number.

sMM.M Example: -02.4

Range: -05.5 to 20.0  
Signed magnitude value.

NNN Example: 134

Range: 000 to 200  
Three digit object size (minutes).

DD\* Example: 56\*

Range: 00\* to 90\*  
Higher' parameter (degrees).

TT.T Example: 59.2

Range: 56.4 to 60.1  
Tracking 'frequency'. Info

(missing command set format line in manual)

Example: CNGC1 976 SU DNEB MAG 3.9 SZ 66.0'  
Range: n/a  
Object information.

Ok Example: 1

Range: 0 or 1  
Status value returned after setting values. If the value is legal 1 is returned, otherwise 0 is returned.

### b. General Telescope Information

Command ACK (ASCII 6)

Returns A, L, P or G  
Gets alignment status, A for alt-azm, L for land, P for polar, G for German mount polar.

Command :GR#

Returns +HH:MM.T#  
Gets the current Right Ascension.

A Mapug-Astronomy post indicated that for the 3.34L ROMS the format is +HH:MM:SS# for RA  
And the format for Declination is sDD\*MM'SS#. Thus adding seconds and arc seconds.

Command :GD#

Returns sDD\*MM#  
Gets the current declination.

Command :GA#

Returns sDD\*MM#  
Gets the current altitude.

Command :GZ#

Returns DDD\*MM#  
Gets the current azimuth.

Command :GS#

Returns HH:MM:SS#  
Gets the current sidereal time.

Command :SS HH:MM:SS#

Returns Ok  
Gets the sidereal time.

Command :GL# :Ga#

Returns HH:MM:SS#  
Gets the local time either in 24 hour (GL) or 12 hour (Ga) format.

Command :SL HH:MM:SS#

Returns Ok  
Sets the local time. NOTE: The parameter should always be in 24 hour format.

Command :GC#

Returns MM/DD/YY#  
Gets the calendar date.

Command :SC MM/DD/YY#

Returns Ok (see description)  
Sets the calendar date.

NOTE: After the Ok, if the date is valid, two strings will be sent. The first will contain the message "Updating planetary data," the second (sent after the planetary calculations) will contain only blanks. Both strings will be terminated by the (\*) symbol.

Command :Gt#

Returns sDD\*MM#  
Gets the latitude of the currently selected site.

Command :St sDD\*MM#

Returns Ok  
Sets the latitude of the currently selected site.

Command :Gg#

Returns DDD\*MM#  
Gets the longitude of the currently selected site.

Command :Sg DDD\*MM#

Returns Ok  
Sets the longitude of the currently selected site.

Command :GG#

Returns sHH#  
Gets the offset from Greenwich Mean Time.

Command :SG sHH#

Returns Ok  
Sets the offset from Greenwich Mean Time.

Command :W1# :W2# :W3# :W4#

Returns Nothing  
Sets the current site number.

### c. Telescope Motion

Command :Mn# :Ms# :Me# :Mw#

Returns Nothing  
Starts motion in the specified direction at the current rate.

Command :MS#

Returns 0, 1, or 2 (see description)  
Slews telescope to current object coordinates. 0 is returned if the telescope can complete the slew, 1 is returned if the object is below the horizon, and 2 is returned if the object is below the 'higher'

limit. If 1 or 2 is returned, a string containing an appropriate message is also returned.

Command :Qn# :Qs# :Qe# :Qw#  
Returns Nothing  
Stops motion in the specified direction. Also stops the telescope if a slew to object is in progress.

Command :Q#  
Returns Nothing  
Stops a slew to an object.

Command :RG# :RC# :RM# :RS#  
Returns Nothing  
Sets the motion rate to guide (RG), center (RC), find (RM), or slew (RS).

**d. Library/Objects**

Command :Gr#  
Returns HH:MM.T#  
Gets object Right Ascension

Command :Sr HH:MM.T#  
Returns Ok  
Sets object Right Ascension.

A Mapug-Astronomy post indicated that for the 3.34L ROMS the format is +HH:MM:SS# for RA  
And the format for Declination is sDD\*MM'SS#. Thus adding seconds and arc seconds.

Command :Gd#  
Returns sDD\*MM#  
Gets object Declination.

Command :Sd sDD\*MM#  
Returns Ok  
Sets object Declination.

Command :CM#  
Returns (see description)  
Sync. Matches current telescope coordinates to the object coordinates and sends a string indicating which object's coordinates were used.

Command :Gy#  
Returns GPDco#  
Gets the 'type' string for the FIND operation. A capital letter means that the corresponding type is selected while a lower case letter indicates it is not.

Command :Sy GPDco#  
Returns Ok  
Gets the %type' string for the FIND operation.

Command :Gq#  
Returns SU#, EX#, VG#, GD#, FR#, PR# or VP#  
Gets the current minimum quality for the FIND operation

Command :Sq#  
Returns Nothing  
Steps to the next minimum quality for the FIND operation.

Command :Gh#  
Returns DD\*#  
Gets the current 'higher' limit.

Command :Sh DD#  
Returns Ok  
Sets the current 'higher' limit.

Command :Gb# :Gf#  
Returns sMM.M#  
Gets the brighter (Gb) or fainter (Gf) magnitude limit for the FIND operation.

Command :Sb sMM.M# :Sf sMM.M#  
Returns Ok  
Sets the brighter (Gb) or fainter (Gf) magnitude limit for the FIND operation.

Command :GI# :Gs#  
Returns NNN\*  
Gets the larger (GI) or smaller (Gs) size limit for the FIND operation.

Command :SI NNN# :Ss NNN#  
Returns Ok  
Sets the larger (GI) or smaller (Gs) size limit for the FIND operation.

Command :GF#  
Returns NNN#  
Gets the field radius of the FIELD operation.

Command :SF NNN#  
Returns Ok  
Sets the field radius of the FIELD operation.

Command :LF#  
Returns Nothing  
Starts a FIND operation.

Command :LN#  
Returns Nothing  
Finds the next object in a FIND sequence.

Command :LB#  
Returns Nothing  
Finds the previous object in a FIND sequence.

Command :Lf#  
Returns (see description)  
Performs a FIELD operation returning a string containing the number of objects in the field and the object that is closest to the center of the field.

Command :LC NNNN# :LM NNNN# :LS NNNN#  
Returns Nothing  
Sets the object to the CNGC (LC), Messier (LM), or Star (LS) specified by the number. Planets are %stars' 901-909.

Command :LI#  
Returns Object Information  
Gets the current object information.

**e. Miscellaneous**

Command :B+# :B-# :B0# :B1# :B2# :B3#  
Returns Nothing  
Increases (B+) or decreases (B-) reticle brightness, or sets to one of the flashing modes (B0, B1, B2 or B3).

Command :F+# :F-# :FQ# :FF# :FS#  
Returns Nothing  
Starts focus out (F+), starts focus in (F-), stops focus change (FQ), sets focus fast (FF) or focus slow (FS)

Command :GM# :GN# :GO# :GP#  
Returns XYZ#  
Gets site name (XYZ). M through P correspond to 1 through 4.

Command :SM XYZ# :SN XYZ# :SO XYZ# :SP XYZ#  
Returns Ok  
Sets site name.

Command :GT#  
Returns TT.T#  
Gets the current track 'frequency'.

Command :ST TT.T#  
Returns Ok  
Sets the current track 'frequency'.

Command :TM# :TQ# :T+# :T-#  
Returns Nothing  
Switch to manual (TM) or quartz (TM). Increment (T+) or decrement (T-) manual frequency by one tenth.

Command :Gc#  
Returns (12) or (24)  
Get 12/24 hour status of clock.

Command :H#  
Returns Nothing  
Toggle 12/24 hour mode.

Command :P#  
Returns "HIGH PRECISION" when ON "LOW PRECISION" when OFF  
Toggles the High Precision Mode ON or OFF.

Command :U#  
Returns Nothing  
Toggles the long format ON or OFF.  
When the long format is active, whenever a request to send or receive position data, the following format is used:

HH:MM:SS

Example: 05:47:45

Range: 00:00.0 to 23:59:59

Hours, minutes, and seconds.

sDD\*MM:SS

Example: +45\*59:45

Range: -90\*00 to +90\*00

Signed degrees, minutes, and seconds (the '\*' represents ASCII 223 which appears on

the handbox as a degree symbol).

DDD\*MM:SS

Example: 254\*09:45

Range: 000\*00 to 359\*59:59

Unsigned degrees, minutes, and seconds.

Command :Lo N#

Returns Ok

Sets the NGC object library. 0 is the NGC library, 1 is the IC library and 2 is the UGC library. This operation

is successful only if the user has a version of the software that includes the desired library.

Command :Ls N#

Returns Ok

Sets the STAR object library type. 0 is the STAR library; 1 is the SAO library, and 2 is the GCVS library. This operation is successful only if the user has a version of the software that includes the desired library.

#### f. Keypad Hand Controller Specific

Command :D#

Returns (see description)

Gets the distance 'bars' string.

Command :\$Q (1-5)#

Returns Nothing

Toggles Smart Drive status.

Command :?#

Command :?+ #

Command :?- #

Returns Page of Help Information

Starts (??) or moves through (?+ or ?-) Help.

Command : G0#

Command : G1#

Command : G2#

Returns Alignment Menu Entry

Used to implement alignment menu

#### LX200 Demo Program

The RS-232 interface communicates with your computer at 9600 Baud Rate, Parity = None, 8 Data Bits, 1 Stop Bit.

For those who are familiar with programming, the LX200 Command Set is written in ASCII character format and

can be used to write your own programs. The LX200 Demo Program is written in Quick Basic and is intended to

demonstrate how commands can be sent to the telescope and information received from the telescope. It is not a

"polished" program and does not incorporate all of the RS-232 features available. The program is set-up to operate

on serial port 2 (COM2). To operate on serial port 1 (COM1) line 4 should be changed from "COM2 to "COM1.

The program is as follows: Please note that Meade Instruments does not support these programs, or programs that you may write in any way. For questions relating to after-market software programs, refer back to those manufacturers.

#### LX200 Test Program

Once you have the RS-232 cable constructed you will want to test the cable. Below is a simple program called "LX200

TEST" that is written in GW Basic programming language and will work with IBM compatible computers. LX200 TEST

is an effective program to fully check the RS-232 line communications from your personal computer to the LX200,

allowing you to concentrate on de-bugging your RS-232 cable.

To enter the following program, first load BASIC or GWBASIC (which ever your computer system uses), then type in the following program. When complete, be sure to save the program as "LX200TST.BAS."

```
10 CLS
```

```
20 DEFINT A-X
```

```
30 OPEN "COM1:9600,N,8,1,CDO,CSO,DSO,RS," FOR RANDOM AS #1
```

```
50 key1$ = INKEY$: IF key1$ = " * " THEN GOTO 50
```

```
60 REM Key 1 loop
```

```
70 IF key1$ = CHR$(119) THEN GOSUB 200: REM "w" key
```

```
80 IF key1$ = CHR$(101) THEN GOSUB 200: REM "e" key
```

```
90 IF key1$ = CHR$(110) THEN GOSUB 200: REM "n" key
```

```
100 IF key1$ = CHR$(115) THEN GOSUB 200: REM "s" key
```

```
105 IF key1$ = "x" THEN END: REM To exit test.
```

```
110 GOTO 50 REM return to Key 1 loop
```

```
120 END
```

```
200 REM.Key 1 direction printouts to COM 1
```

```
210 REM west
```

```
220 IF key1$ = "w" THEN a$ = " #:Mw#" : PRINT #1, a$: REM GO west
```

```
230 REM east
```

```
240 IF key1$ = "e" THEN a$ = " #:Me#" : PRINT #1, a$: REM GO east
```

```
250 REM north
```

```
260 IF key1$ = "n" THEN a$ = " #:Mn#" : PRINT #1, a$: REM GO north
```

```
270 REM south:
```

```
280 IF key1$ = "s" THEN a$ = " #:Ms#" : PRINT #1, a$: REM GO south
```

```
290 key1$ = INKEY$:
```

```
300 IF key1$ = CHR$(32) THEN GOTO 400 ELSE GOTO 200
```

```
400 REM This stops motion (by hitting SPACE bar).
```

```
410 B$ = " #:Qe#" : PRINT #1, B$
```

```
420 B$ = " #:Qw#" : PRINT #1, B$
```

```
430 B$ = " #:Qn#" : PRINT #1, B$
```

```
440 B$ = " #:Qs#" : PRINT #1, B$
```

```
450 RETURN
```

```
460 END
```

To use the above program, connect the completed cable to your PC serial port and to the LX200 RS-232 Port. Load BASIC (or GWBASIC), if not already loaded, and run "LX200TST.BAS - " Nothing will appear on the computer screen. Press any one of the N, S, E, or W (lower case) keys on your PC keyboard, this will move the LX200 North, South, East, or West respectively. Press the space bar on the PC keyboard to stop. Press X to exit the program. If the LX200 does not respond to the N, S, E, or W keys, be sure the CAPSLOCK is OFF. If it still does not work, check the PC serial port pinouts for your computer to be sure they are wired correctly to the LX200 6 line connector. With a successful check-out of the PC link with the LX200 using "TEST", you are now ready to write your own software program using the LX200 Command Set or to use the sample program called "DEMO" that is written in Quick Basic software language.

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